

DSC 411/511
Advance Data Mining Algorithms and Applications

Fall 2026

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Location: MSB 104



Lecture 11

Association Analysis (Apriori and FP-Growth)

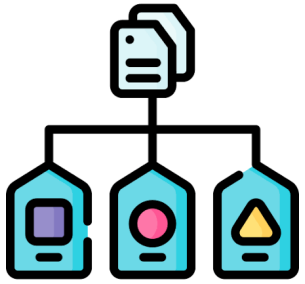
Today's class:

- **Apriori**
- **FP-Growth**

What We Have Covered So Far

Classification

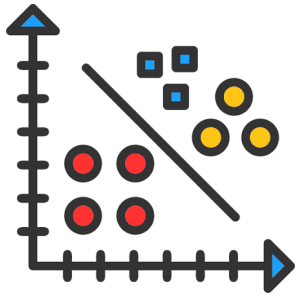
which category does this row belong to?



- Decision trees
- naive Bayes
- k-nearest neighbors
- logistic regression
- support vector machines

Clustering

which rows are similar to each other?



- K-mean clustering
- Hierarchical
- DBSCAN.

Association

Which things tend to appear together?



- Apriori association
- FP-Growth

Which Products Travel Together?

The problem

The Question

- Which two items keep turning up together?
- If somebody buys bread and butter, what else is likely in the basket?
- Which pair would surprise you?

50,000 products and 2 million receipts:

- You found that pair in three seconds, by eye.
- Nobody can do that for 2 million receipts.
- And the interesting pairs are rare. A combination appearing in 2 percent of baskets is still 40,000 receipts, and you would never spot it by scrolling.



15/03/2024 02:00
Cashier: Lisa M. Register: 07

3x Organic Bananas (PLU 4011)	$9
2x Whole Wheat Bread	$16
2% Milk - Gallon	$4
Fresh Ground Coffee 12oz	$10
Loyalty Card Savings	$-4
Subtotal:	$25
Tax:	$2
You Saved Today:	$3.50
Total:	**$24**
Card:	•••• 4892
Type:	Debit
Entry:	Chip
Time:	10/10/2025, 18:30:55
Ref:	AUTH-998877
Status:	Approved



SUPERMARKET
465 S Arroyo Pkwy
Pasadena, CA 91105

1	DOZ FREE RANGE EGG	$3.89
1	GV ORGANIC MILK 2%	$4.19
1	SBUX FRENCH ROAST WL	$8.99
SUBTOTAL		$17.07
TOTAL		$17.07
		$17.07
PURCHASE		$17.07
		VISA DEBIT 3991
Auth #140987		Exp Date **/**
Lane #14		Cashier 1409
11/21/2019 01:28 PM		Ref/Seq #140987
MRCH 204650		Term=001 IC=CC
EPS Sequence		934518
ITEM 1		H,STEPHIE
11/21/2019	01:28 PM	4130 29 1041 2675



Walmart
Save money. Live better.

(813) 932-0562
Manager COLLEEN BRICKEY
8885 N FLORIDA AVE
TAMPA FL 33604

ST# 5221 OP# 00001061 TE# 06 TR# 05332

BREAD	007225003712	F	2.88 N
BREAD	007225003712	F	2.88 N
GV PNT BUTTR	007874237003	F	3.84 N
GV PNT BUTTR	007874237003	F	3.84 N
GV PNT BUTTR	007874237003	F	3.84 N
GV PNT BUTTR	007874237003	F	3.84 N
GV PARM 16OZ	007874201510	F	4.98 O
GV CHNK CHKN	007874206784	F	1.98 N
GV CHNK CHKN	007874206784	F	1.98 N
12 CT NITRIL	073191913822		2.78 X
FOLGERS	002550000377	F	10.48 N
SC TWIST UP	007874222682	F	0.84 X
EGGS	060538871459	F	1.88 O
	SUBTOTAL		46.04
TAX 1	7.000 %		0.26
	TOTAL		46.30
	DEBIT TEND		46.30
	CHANGE DUE		0.00

EFT DEBIT PAY FROM PRIMARY
ACCOUNT : 5269
46.30 TOTAL PURCHASE
PAYMENT DECLINED DEBIT NOT AVAILABLE
11/06/11 02:21:54

EFT DEBIT PAY FROM PRIMARY
ACCOUNT : 5269
46.30 TOTAL PURCHASE
REF # 131000195280
NETWORK ID. 0071 APPR CODE 297664
11/06/11 02:22:54

ITEMS SOLD 13

TC# 0432 2121 1542 2401 9590

What Association Analysis Is

Definition

Association analysis discovers relationships, expressed as association rules or sets of frequent items, hidden in large data sets.

Chapter 5
Introduction to Data Mining, 2nd Edition

The output is a rule

- baskets containing bread and butter tend to also contain milk.
  {bread, butter} → {milk}

	Ask	Example
Support	How often does this combination appear at all?	3 of 100 baskets, so 3%
Confidence	When the left side appears, how often does the right side follow?	3 of 4 times, so 75%

This is unsupervised, like clustering.
No labels, no accuracy, no answer key.

What Association Analysis Is

The Combinatorial Problem

Just count every combination, surely?

- 5 items gives 31 possible combinations.
- 20 items gives about 1 million.
- 100 items gives more combinations than there are atoms in the observable universe.

Where it is used

- Retail baskets, the original application
- Web pages viewed in the same session
- Symptoms recorded in the same patient
- Faults appearing together in the same machine

What we need

- A way to **skip** the combinations that cannot possibly be frequent.
- Without counting them first.
- Two ways to skip combinations that cannot possibly be frequent, Apriori & FP-Growth.

The Apriori Principle

The Principle

If an itemset is frequent, then every subset of it must also be frequent.

Turn it around, and it becomes useful

If an itemset is **not** frequent, then no superset of it can be frequent either.

Why that is obvious once you see it

- {bread, milk} appears in 10 baskets.
- {bread, milk, eggs} can appear in at most 10. Never more.
- So if {bread, milk} fell below the threshold, {bread, milk, eggs} is already dead.

What it gives you

- The moment a combination fails, you discard every combination containing it.
- One failed pair can eliminate thousands of larger sets you never have to count.

Apriori

Definition

Apriori generates frequent itemset level by level, using the frequent itemset of size k to build the candidates of size k+1, and pruning any candidate with an infrequent subset.

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How it works

1. **C1:** count the support of every single item
2. **L1:** keep the ones that pass minimum support
3. **Ck:** join L(k-1) with itself to build candidates, and prune any with an infrequent subset
4. **Lk:** scan, count, keep the ones that pass
5. Verify the generated rules

Pros

- Simple to explain and simple to implement.
- The pruning step is easy to see and easy to check by hand.

Cons

- It scans the entire dataset once per level. Ten levels means ten passes.
- Candidate generation explodes when items are numerous or support is low.

Apriori

Definition

Apriori generates frequent itemset level by level, using the frequent itemset of size k to build the candidates of size k+1, and pruning any candidate with an infrequent subset.

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ID	Items
1	Bread, Butter, Jam, Milk
2	Bread, Butter, Milk
3	Bread, Juice, Curd
4	Bread, Milk, Juice
5	Butter, Milk, Juice

Support: 50%
Confident: 75%

Generate rules to see which basket will have what items?

Apriori

Definition

Apriori generates frequent itemset level by level, using the frequent itemset of size k to build the candidates of size k+1, and pruning any candidate with an infrequent subset.

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Step1: count the support of every single item (**C1**)

ID	Items
1	Bread, Butter, Jam, Milk
2	Bread, Butter, Milk
3	Bread, Juice, Curd
4	Bread, Milk, Juice
5	Butter, Milk, Juice

Items	Count
Bread	4
Butter	
Jam	
Milk	
Juice	
Curd	

Support: 50%
Confident: 75%

Apriori

Definition

Apriori generates frequent itemset level by level, using the frequent itemset of size k to build the candidates of size k+1, and pruning any candidate with an infrequent subset.

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Step1: count the support of every single item (**C1**)

ID	Items
1	Bread, Butter, Jam, Milk
2	Bread, Butter, Milk
3	Bread, Juice, Curd
4	Bread, Milk, Juice
5	Butter, Milk, Juice

Items	Count
Bread	4
Butter	3
Jam	
Milk	
Juice	
Curd	

Support: 50%
Confident: 75%

Apriori

Definition

Apriori generates frequent itemset level by level, using the frequent itemset of size k to build the candidates of size k+1, and pruning any candidate with an infrequent subset.

Chapter 5
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Step1: count the support of every single item (**C1**)

ID	Items
1	Bread, Butter, Jam, Milk
2	Bread, Butter, Milk
3	Bread, Juice, Curd
4	Bread, Milk, Juice
5	Butter, Milk, Juice

Items	Count
Bread	4
Butter	3
Jam	1
Milk	4
Juice	3
Curd	1

Support: 50%
Confident: 75%

Apriori

Definition

Apriori generates frequent itemset level by level, using the frequent itemset of size k to build the candidates of size k+1, and pruning any candidate with an infrequent subset.

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Step2: keep the ones that pass minimum support (**L1**)

ID	Items
1	Bread, Butter, Jam, Milk
2	Bread, Butter, Milk
3	Bread, Juice, Curd
4	Bread, Milk, Juice
5	Butter, Milk, Juice

Items	Count	Support
Bread	4	
Butter	3	
Jam	1	
Milk	4	
Juice	3	
Curd	1	

Count/ Total Transaction

Support: 50%
Confident: 75%

Apriori

Definition

Apriori generates frequent itemset level by level, using the frequent itemset of size k to build the candidates of size k+1, and pruning any candidate with an infrequent subset.

Chapter 5
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Step2: keep the ones that pass minimum support (**L1**)

ID	Items
1	Bread, Butter, Jam, Milk
2	Bread, Butter, Milk
3	Bread, Juice, Curd
4	Bread, Milk, Juice
5	Butter, Milk, Juice

Items	Count	Support
Bread	4	4/5
Butter	3	3/5
Jam	1	1/5
Milk	4	4/5
Juice	3	3/5
Curd	1	1/5

Count/ Total Transaction

Support: 50%
Confident: 75%

Apriori

Definition

Apriori generates frequent itemset level by level, using the frequent itemset of size k to build the candidates of size k+1, and pruning any candidate with an infrequent subset.

Chapter 5
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Step2: keep the ones that pass minimum support (**L1**)

ID	Items
1	Bread, Butter, Jam, Milk
2	Bread, Butter, Milk
3	Bread, Juice, Curd
4	Bread, Milk, Juice
5	Butter, Milk, Juice

Items	Count	Support
Bread	4	0.8
Butter	3	0.6
Jam	1	0.2
Milk	4	0.8
Juice	3	0.6
Curd	1	0.2

Count/ Total Transaction

Support: 50%
Confident: 75%

Apriori

Definition

Apriori generates frequent itemset level by level, using the frequent itemset of size k to build the candidates of size k+1, and pruning any candidate with an infrequent subset.

Chapter 5
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Step2: keep the ones that pass minimum support (**L1**)

ID	Items
1	Bread, Butter, Jam, Milk
2	Bread, Butter, Milk
3	Bread, Juice, Curd
4	Bread, Milk, Juice
5	Butter, Milk, Juice

Items	Count	Support (%)
Bread	4	80
Butter	3	60
~~Jam~~	~~1~~	~~20~~
Milk	4	80
Juice	3	60
~~Curd~~	~~1~~	~~20~~

Support: 50%
Confident: 75%

Apriori

Definition

Apriori generates frequent itemset level by level, using the frequent itemset of size k to build the candidates of size k+1, and pruning any candidate with an infrequent subset.

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Step2: keep the ones that pass minimum support (**L1**)

ID	Items
1	Bread, Butter, Jam, Milk
2	Bread, Butter, Milk
3	Bread, Juice, Curd
4	Bread, Milk, Juice
5	Butter, Milk, Juice

Items	Count	Support (%)
Bread	4	80
Butter	3	60
~~Jam~~	~~1~~	~~20~~
Milk	4	80
Juice	3	60
~~Curd~~	~~1~~	~~20~~

→ Bread
Butter
Milk
Juice

Support: 50% ←
Confident: 75%

Apriori

Definition

Apriori generates frequent itemset level by level, using the frequent itemset of size k to build the candidates of size k+1, and pruning any candidate with an infrequent subset.

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Step3: join L(k-1) with itself to build candidates, and prune any with an infrequent subset

ID	Items
1	Bread, Butter, Jam, Milk
2	Bread, Butter, Milk
3	Bread, Juice, Curd
4	Bread, Milk, Juice
5	Butter, Milk, Juice

Bread
Butter
Milk
Juice

Support: 50%
Confident: 75%

Apriori

Definition

Apriori generates frequent itemset level by level, using the frequent itemset of size k to build the candidates of size k+1, and pruning any candidate with an infrequent subset.

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Step3: join L(k-1) with itself to build candidates, and prune any with an infrequent subset

ID	Items
1	Bread, Butter, Jam, Milk
2	Bread, Butter, Milk
3	Bread, Juice, Curd
4	Bread, Milk, Juice
5	Butter, Milk, Juice

Support: 50%
Confident: 75%

Items	Count	Support (%)
Bread, Butter		
Bread, Milk		
Bread, Juice		
Butter, Milk		
Butter, Juice		
Milk, Juice		

Bread
Butter
Milk
Juice

Apriori

Definition

Apriori generates frequent itemset level by level, using the frequent itemset of size k to build the candidates of size k+1, and pruning any candidate with an infrequent subset.

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Step3: join L(k-1) with itself to build candidates, and prune any with an infrequent subset

ID	Items
1	Bread, Butter, Jam, Milk
2	Bread, Butter, Milk
3	Bread, Juice, Curd
4	Bread, Milk, Juice
5	Butter, Milk, Juice

Items	Count	Support (%)
Bread, Butter	2	
Bread, Milk		
Bread, Juice		
Butter, Milk		
Butter, Juice		
Milk, Juice		

Support: 50%
Confident: 75%

Apriori

Definition

Apriori generates frequent itemset level by level, using the frequent itemset of size k to build the candidates of size k+1, and pruning any candidate with an infrequent subset.

Chapter 5
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Step3: join L(k-1) with itself to build candidates, and prune any with an infrequent subset

ID	Items
1	Bread, Butter, Jam, Milk
2	Bread, Butter, Milk
3	Bread, Juice, Curd
4	Bread, Milk, Juice
5	Butter, Milk, Juice

Items	Count	Support (%)
Bread, Butter	2	
Bread, Milk	3	
Bread, Juice		
Butter, Milk		
Butter, Juice		
Milk, Juice		

Support: 50%
Confident: 75%

Apriori

Definition

Apriori generates frequent itemset level by level, using the frequent itemset of size k to build the candidates of size k+1, and pruning any candidate with an infrequent subset.

Chapter 5
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Step3: join L(k-1) with itself to build candidates, and prune any with an infrequent subset

ID	Items
1	Bread, Butter, Jam, Milk
2	Bread, Butter, Milk
3	Bread, Juice, Curd
4	Bread, Milk, Juice
5	Butter, Milk, Juice

Items	Count	Support (%)
Bread, Butter	2	
Bread, Milk	3	
Bread, Juice	2	
Butter, Milk	3	
Butter, Juice	1	
Milk, Juice	2	

Support: 50%
Confident: 75%

Apriori

Definition

Apriori generates frequent itemset level by level, using the frequent itemset of size k to build the candidates of size k+1, and pruning any candidate with an infrequent subset.

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Step3: join L(k-1) with itself to build candidates, and prune any with an infrequent subset

ID	Items
1	Bread, Butter, Jam, Milk
2	Bread, Butter, Milk
3	Bread, Juice, Curd
4	Bread, Milk, Juice
5	Butter, Milk, Juice

Items	Count	Support (%)
~~Bread, Butter~~	~~2~~	~~40~~
Bread, Milk	3	60
~~Bread, Juice~~	~~2~~	~~40~~
Butter, Milk	3	60
~~Butter, Juice~~	~~1~~	~~20~~
~~Milk, Juice~~	~~2~~	~~40~~

Support: 50% ←
Confident: 75%

Apriori

Definition

Apriori generates frequent itemset level by level, using the frequent itemset of size k to build the candidates of size k+1, and pruning any candidate with an infrequent subset.

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Step3: join L(k-1) with itself to build candidates, and prune any with an infrequent subset

ID	Items
1	Bread, Butter, Jam, Milk
2	Bread, Butter, Milk
3	Bread, Juice, Curd
4	Bread, Milk, Juice
5	Butter, Milk, Juice

Support: 50% ←
Confident: 75%

Items	Count	Support (%)
Bread, Butter	2	40
Bread, Milk	3	60
Bread, Juice	2	40
Butter, Milk	3	60
Butter, Juice	1	20
Milk, Juice	2	40

→ {Bread, Milk}
{Butter, Milk}

Apriori

Definition

Apriori generates frequent itemset level by level, using the frequent itemset of size k to build the candidates of size k+1, and pruning any candidate with an infrequent subset.

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Step4: scan, count, keep the ones that pass

ID	Items
1	Bread, Butter, Jam, Milk
2	Bread, Butter, Milk
3	Bread, Juice, Curd
4	Bread, Milk, Juice
5	Butter, Milk, Juice

{Bread, Milk}
{Butter, Milk}
↓
{Bread, Butter, Milk}

Support: 50%
Confident: 75%

Apriori

Definition

Apriori generates frequent itemset level by level, using the frequent itemset of size k to build the candidates of size k+1, and pruning any candidate with an infrequent subset.

Chapter 5
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Step5: Verify the generated rules

ID	Items
1	Bread, Butter, Jam, Milk
2	Bread, Butter, Milk
3	Bread, Juice, Curd
4	Bread, Milk, Juice
5	Butter, Milk, Juice

{Bread, Milk}

Conf (Bread → Milk}

Conf (Milk → Bread}

{Butter, Milk}

Conf (Butter → Milk}

Conf (Milk → Butter}

$$\text{Conf}(x \rightarrow y) = \frac{S(x \cup y)}{S(x)}$$

Support: 50%
Confident: 75%

Apriori

Definition

Apriori generates frequent itemset level by level, using the frequent itemset of size k to build the candidates of size k+1, and pruning any candidate with an infrequent subset.

Chapter 5
Introduction to Data Mining, 2nd Edition

Step5: Verify the generated rules

$$\text{Conf}(x \rightarrow y) = \frac{S(x \cup y)}{S(x)}$$

ID	Items
1	Bread, Butter, Jam, Milk
2	Bread, Butter, Milk
3	Bread, Juice, Curd
4	Bread, Milk, Juice
5	Butter, Milk, Juice

{Bread, Milk}

Conf (Bread → Milk}

Conf (Milk → Bread}

{Butter, Milk}

Conf (Butter → Milk}

Conf (Milk → Butter}

Items	S(%)
Bread	80
Milk	80
Butter	60
Bread, Milk	60
Butter, Milk	60

Support: 50%
Confident: 75%

Apriori

Definition

Apriori generates frequent itemset level by level, using the frequent itemset of size k to build the candidates of size k+1, and pruning any candidate with an infrequent subset.

Chapter 5
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Step5: Verify the generated rules

$$\text{Conf}(x \rightarrow y) = \frac{S(x \cup y)}{S(x)}$$

ID	Items
1	Bread, Butter, Jam, Milk
2	Bread, Butter, Milk
3	Bread, Juice, Curd
4	Bread, Milk, Juice
5	Butter, Milk, Juice

{Bread, Milk}

Conf (Bread → Milk} = 60/80

Conf (Milk → Bread} = 60/80

{Butter, Milk}

Conf (Butter → Milk} = 60/60

Conf (Milk → Butter} = 60/80

Items	S(%)
Bread	80
Milk	80
Butter	60
Bread, Milk	60
Butter, Milk	60

Support: 50%
Confident: 75%

Apriori

Definition

Apriori generates frequent itemset level by level, using the frequent itemset of size k to build the candidates of size k+1, and pruning any candidate with an infrequent subset.

Chapter 5
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Step5: Verify the generated rules

$$\text{Conf}(x \rightarrow y) = \frac{S(x \cup y)}{S(x)}$$

ID	Items
1	Bread, Butter, Jam, Milk
2	Bread, Butter, Milk
3	Bread, Juice, Curd
4	Bread, Milk, Juice
5	Butter, Milk, Juice

{Bread, Milk}

Conf (Bread → Milk} = 60/80 = 75%

Conf (Milk → Bread} = 60/80 = 75%

{Butter, Milk}

Conf (Butter → Milk} = 60/60 = 100%

Conf (Milk → Butter} = 60/80 = 75%

Items	S(%)
Bread	80
Milk	80
Butter	60
Bread, Milk	60
Butter, Milk	60

Support: 50%
Confident: 75%

Apriori

Definition

Apriori generates frequent itemset level by level, using the frequent itemset of size k to build the candidates of size k+1, and pruning any candidate with an infrequent subset.

Chapter 5
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Step5: Verify the generated rules

ID	Items
1	Bread, Butter, Jam, Milk
2	Bread, Butter, Milk
3	Bread, Juice, Curd
4	Bread, Milk, Juice
5	Butter, Milk, Juice

{Bread, Milk}

Conf (Bread → Milk} = 75%

Conf (Milk → Bread} = 75%

{Butter, Milk}

Conf (Butter → Milk} = 100%

Conf (Milk → Butter} = 75%

Support: 50%
Confident: 75%

Frequent Pattern Growth (FP-Growth)

Definition

FP-Growth compresses the data set into a frequent pattern tree, then mines frequent itemsets directly from that tree without generating candidates.

Chapter 5
Introduction to Data Mining, 2nd Edition

How it works

1. Count every single item once. Discard the infrequent ones.
2. Sort each transaction by item frequency.
3. Insert every transaction into a tree, sharing prefixes where they match.

Pros

- Much faster than Apriori on large datasets, often by an order of magnitude.
- Only two scans of the data, regardless of how long the itemset get.

Cons

- Harder to follow by hand.
- Must fit in memory.

Frequent Pattern Growth (FP-Growth)

Definition

FP-Growth compresses the data set into a frequent pattern tree, then mines frequent itemsets directly from that tree without generating candidates.

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ID	Items
1	S, Q
2	R, Q, T, P, S
3	S
4	P, S, T
5	R, P, Q, T
6	T, P, S, Q
7	Q, P, S
8	Q, R, U
9	U

Step1: Count every single item once. Discard the infrequent ones.

Support: 3

Frequent Pattern Growth (FP-Growth)

Definition

FP-Growth compresses the data set into a frequent pattern tree, then mines frequent itemsets directly from that tree without generating candidates.

Chapter 5
Introduction to Data Mining, 2nd Edition

ID	Items
1	S, Q
2	R, Q, T, P, S
3	S
4	P, S, T
5	R, P, Q, T
6	T, P, S, Q
7	Q, P, S
8	Q, R, U
9	U

Support: 3

Step1: Count every single item once. Discard the infrequent ones.

Item	Freq
P	5
Q	6
R	3
S	6
T	4
U	2

Frequent Pattern Growth (FP-Growth)

Definition

FP-Growth compresses the data set into a frequent pattern tree, then mines frequent itemsets directly from that tree without generating candidates.

Chapter 5
Introduction to Data Mining, 2nd Edition

ID	Items
1	S, Q
2	R, Q, T, P, S
3	S
4	P, S, T
5	R, P, Q, T
6	T, P, S, Q
7	Q, P, S
8	Q, R, ~~U~~
~~9~~	~~U~~

Support: 3

Step1: Count every single item once. Discard the infrequent ones.

Item	Freq
P	5
Q	6
R	3
S	6
T	4
~~U~~	~~2~~

Frequent Pattern Growth (FP-Growth)

Definition

FP-Growth compresses the data set into a frequent pattern tree, then mines frequent itemsets directly from that tree without generating candidates.

Chapter 5
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ID	Items
1	S, Q
2	R, Q, T, P, S
3	S
4	P, S, T
5	R, P, Q, T
6	T, P, S, Q
7	Q, P, S
8	Q, R,



Step2: Sort each transaction by item frequency.

Item	Freq	Priority
P	5	3
Q	6	1
R	3	5
S	6	2
T	4	4

Q > S > P > T > R

Support: 3

Frequent Pattern Growth (FP-Growth)

Definition

FP-Growth compresses the data set into a frequent pattern tree, then mines frequent itemsets directly from that tree without generating candidates.

ID	Items	Ordered Items
1	S, Q	Q, S
2	R, Q, T, P, S	Q, S, P, T, R
3	S	S
4	P, S, T	S, P, T
5	R, P, Q, T	Q, P, T, R
6	T, P, S, Q	Q, S, P, T
7	Q, P, S	Q, S, P
8	Q, R,	Q, R

Step2: Sort each transaction by item frequency.

Q > S > P > T > R

Support: 3

Frequent Pattern Growth (FP-Growth)

Definition

FP-Growth compresses the data set into a frequent pattern tree, then mines frequent itemsets directly from that tree without generating candidates.

Chapter 5
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ID	Items	Ordered Items
1	S, Q	Q, S
2	R, Q, T, P, S	Q, S, P, T, R
3	S	S
4	P, S, T	S, P, T
5	R, P, Q, T	Q, P, T, R
6	T, P, S, Q	Q, S, P, T
7	Q, P, S	Q, S, P
8	Q, R,	Q, R

Step 3: Insert every transaction into a tree, sharing prefixes where they match.



Support: 3

Frequent Pattern Growth (FP-Growth)

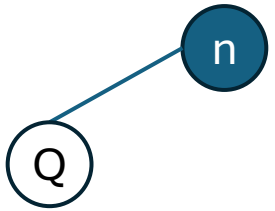
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Chapter 5
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ID	Items	Ordered Items
1	S, Q	Q, S ←
2	R, Q, T, P, S	Q, S, P, T, R
3	S	S
4	P, S, T	S, P, T
5	R, P, Q, T	Q, P, T, R
6	T, P, S, Q	Q, S, P, T
7	Q, P, S	Q, S, P
8	Q, R,	Q, R

Step 3: Insert every transaction into a tree, sharing prefixes where they match.



Q:1

Support: 3

Frequent Pattern Growth (FP-Growth)

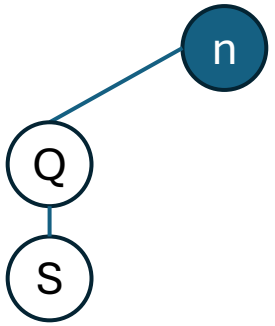
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Chapter 5
Introduction to Data Mining, 2nd Edition

ID	Items	Ordered Items
1	S, Q	Q, S ←
2	R, Q, T, P, S	Q, S, P, T, R
3	S	S
4	P, S, T	S, P, T
5	R, P, Q, T	Q, P, T, R
6	T, P, S, Q	Q, S, P, T
7	Q, P, S	Q, S, P
8	Q, R,	Q, R

Step 3: Insert every transaction into a tree, sharing prefixes where they match.



Q:1
S:1

Support: 3

Frequent Pattern Growth (FP-Growth)

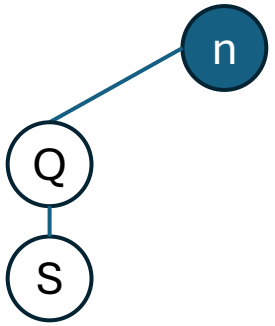
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ID	Items	Ordered Items
1	S, Q	Q, S
2	R, Q, T, P, S	Q, S, P, T, R ←
3	S	S
4	P, S, T	S, P, T
5	R, P, Q, T	Q, P, T, R
6	T, P, S, Q	Q, S, P, T
7	Q, P, S	Q, S, P
8	Q, R,	Q, R

Step 3: Insert every transaction into a tree, sharing prefixes where they match.



Q:1, 2
S:1, 2

Support: 3

Frequent Pattern Growth (FP-Growth)

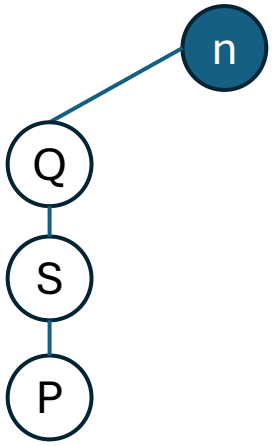
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Introduction to Data Mining, 2nd Edition

ID	Items	Ordered Items
1	S, Q	Q, S
2	R, Q, T, P, S	Q, S, P, T, R ←
3	S	S
4	P, S, T	S, P, T
5	R, P, Q, T	Q, P, T, R
6	T, P, S, Q	Q, S, P, T
7	Q, P, S	Q, S, P
8	Q, R,	Q, R

Step 3: Insert every transaction into a tree, sharing prefixes where they match.



Q:1, 2
S:1, 2
P:1

Support: 3

Frequent Pattern Growth (FP-Growth)

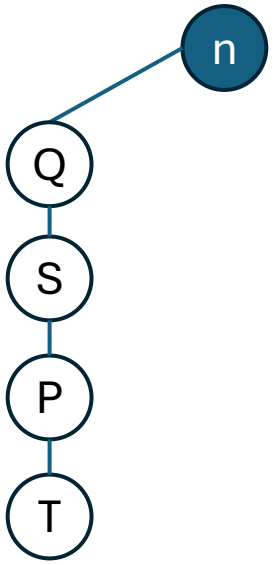
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Introduction to Data Mining, 2nd Edition

ID	Items	Ordered Items
1	S, Q	Q, S
2	R, Q, T, P, S	Q, S, P, T, R ←
3	S	S
4	P, S, T	S, P, T
5	R, P, Q, T	Q, P, T, R
6	T, P, S, Q	Q, S, P, T
7	Q, P, S	Q, S, P
8	Q, R,	Q, R

Step 3: Insert every transaction into a tree, sharing prefixes where they match.



Q:1, 2
S:1, 2
P:1
T:1

Support: 3

Frequent Pattern Growth (FP-Growth)

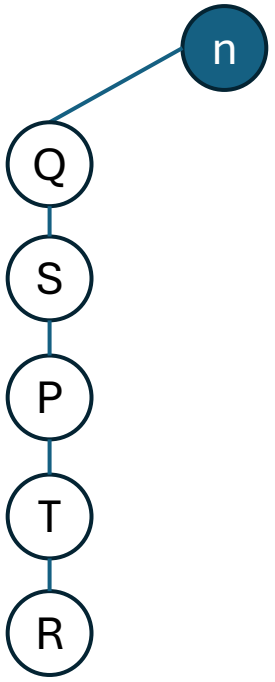
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Introduction to Data Mining, 2nd Edition

ID	Items	Ordered Items
1	S, Q	Q, S
2	R, Q, T, P, S	Q, S, P, T, R ←
3	S	S
4	P, S, T	S, P, T
5	R, P, Q, T	Q, P, T, R
6	T, P, S, Q	Q, S, P, T
7	Q, P, S	Q, S, P
8	Q, R,	Q, R

Step 3: Insert every transaction into a tree, sharing prefixes where they match.



Q:1, 2
S:1, 2
P:1
T:1
R: 1

Support: 3

Frequent Pattern Growth (FP-Growth)

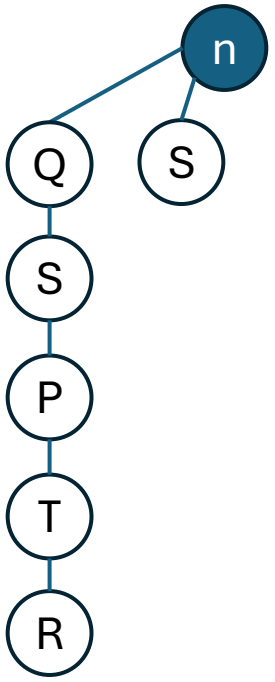
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Chapter 5
Introduction to Data Mining, 2nd Edition

ID	Items	Ordered Items
1	S, Q	Q, S
2	R, Q, T, P, S	Q, S, P, T, R
3	S	S ←
4	P, S, T	S, P, T
5	R, P, Q, T	Q, P, T, R
6	T, P, S, Q	Q, S, P, T
7	Q, P, S	Q, S, P
8	Q, R,	Q, R

Step 3: Insert every transaction into a tree, sharing prefixes where they match.



Q:1, 2
S:1, 2
P:1
T:1
R:1
S:1

Support: 3

Frequent Pattern Growth (FP-Growth)

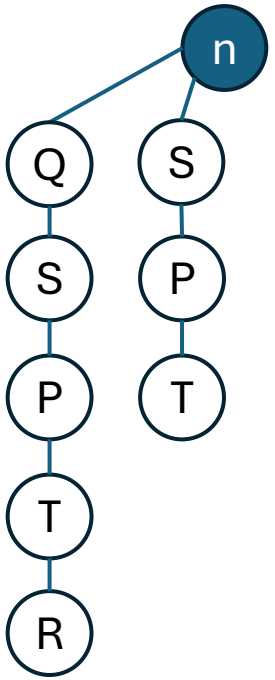
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Chapter 5
Introduction to Data Mining, 2nd Edition

ID	Items	Ordered Items
1	S, Q	Q, S
2	R, Q, T, P, S	Q, S, P, T, R
3	S	S
4	P, S, T	S, P, T ←
5	R, P, Q, T	Q, P, T, R
6	T, P, S, Q	Q, S, P, T
7	Q, P, S	Q, S, P
8	Q, R,	Q, R

Step 3: Insert every transaction into a tree, sharing prefixes where they match.



Q:1, 2
S:1, 2
P:1
T:1
R:1
S:1,2
P:1
T:1

Support: 3

Frequent Pattern Growth (FP-Growth)

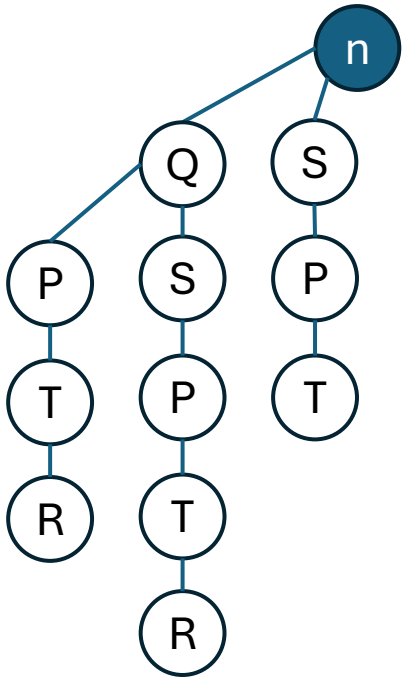
Definition

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Chapter 5
Introduction to Data Mining, 2nd Edition

ID	Items	Ordered Items
1	S, Q	Q, S
2	R, Q, T, P, S	Q, S, P, T, R
3	S	S
4	P, S, T	S, P, T
5	R, P, Q, T	Q, P, T, R ←
6	T, P, S, Q	Q, S, P, T
7	Q, P, S	Q, S, P
8	Q, R,	Q, R

Step 3: Insert every transaction into a tree, sharing prefixes where they match.



Q:1, 2, 3
S:1, 2
P:1
T:1
R:1
S:1,2
P:1
T:1
P:1
T:1
R:1

Support: 3

Frequent Pattern Growth (FP-Growth)

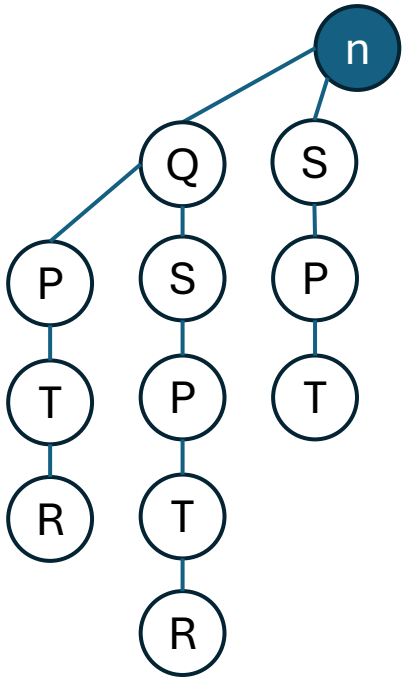
Definition

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Chapter 5
Introduction to Data Mining, 2nd Edition

ID	Items	Ordered Items
1	S, Q	Q, S
2	R, Q, T, P, S	Q, S, P, T, R
3	S	S
4	P, S, T	S, P, T
5	R, P, Q, T	Q, P, T, R
6	T, P, S, Q	Q, S, P, T ←
7	Q, P, S	Q, S, P
8	Q, R,	Q, R

Step 3: Insert every transaction into a tree, sharing prefixes where they match.



Q:1, 2, 3, 4
S:1, 2, 3
P:1, 2
T:1, 2
R:1
S:1,2
P:1
T:1
P:1
T:1
R:1

Support: 3

Frequent Pattern Growth (FP-Growth)

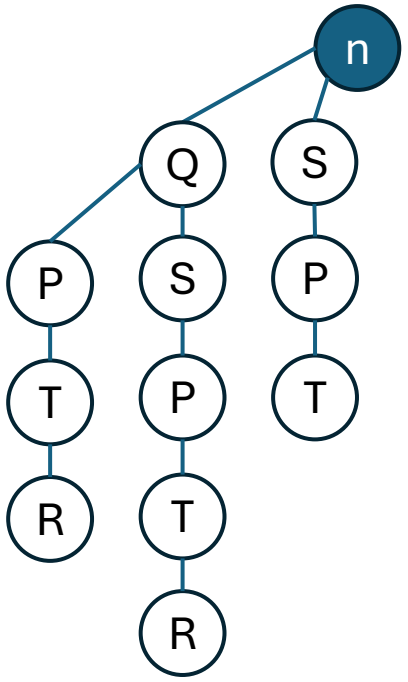
Definition

FP-Growth compresses the data set into a frequent pattern tree, then mines frequent itemsets directly from that tree without generating candidates.

Chapter 5
Introduction to Data Mining, 2nd Edition

ID	Items	Ordered Items
1	S, Q	Q, S
2	R, Q, T, P, S	Q, S, P, T, R
3	S	S
4	P, S, T	S, P, T
5	R, P, Q, T	Q, P, T, R
6	T, P, S, Q	Q, S, P, T
7	Q, P, S	Q, S, P ←
8	Q, R,	Q, R

Step 3: Insert every transaction into a tree, sharing prefixes where they match.



Q:1, 2, 3, 4, 5
S:1, 2, 3, 4
P:1, 2, 3
T:1, 2
R:1
S:1,2
P:1
T:1
P:1
T:1
R:1

Support: 3

Frequent Pattern Growth (FP-Growth)

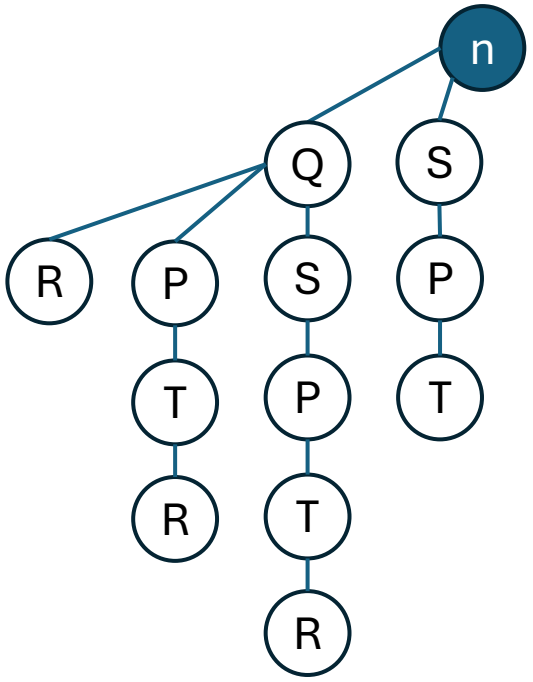
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Chapter 5
Introduction to Data Mining, 2nd Edition

ID	Items	Ordered Items
1	S, Q	Q, S
2	R, Q, T, P, S	Q, S, P, T, R
3	S	S
4	P, S, T	S, P, T
5	R, P, Q, T	Q, P, T, R
6	T, P, S, Q	Q, S, P, T
7	Q, P, S	Q, S, P
8	Q, R,	Q, R ←

Step 3: Insert every transaction into a tree, sharing prefixes where they match.



Q:1, 2, 3, 4, 5, 6
S:1, 2, 3, 4
P:1, 2, 3
T:1, 2
R:1
S:1,2
P:1
T:1
P:1
T:1
R:1
R:1

Support: 3

Association on Ames

Our Question, and What the Code Will Do

- **The question**

  "Given everything we know about these houses, **which features tend to travel together?** And is that new record plausible, or is somebody making it up?"

- **How Association Analysis asks it differently?**

  Classification asked: which category does this **row** belong to?
  Clustering asked: which **rows** are similar to each other?
  Association asks: **which features appear in the same house, more often than chance would predict?**

Rule: IF Large house AND High quality THEN Big garage

support 0.28: 28% of all houses have all three facts
confidence 0.85: of the large, high-quality houses, 85% have a big garage
base rate 0.47: but only 47% of ALL houses have a big garage
lift 1.81: 0.85 / 0.47